

Kenny Nguyen

DATA 205

Professor Perine

Maryland Appeals and Grievances Ingestion/ Cleaning

1. I downloaded the Reports on the Health Care Appeals and Grievance Law from Maryland.gov for the years 2021-2024.
2021: [https://insurance.maryland.gov/Consumer/Appeals and Grievances Reports/2021-Report-on-the-Health-Care-Appeals-and-Grievance-Law.pdf](https://insurance.maryland.gov/Consumer/Appeals%20and%20Grievances%20Reports/2021-Report-on-the-Health-Care-Appeals-and-Grievance-Law.pdf)
2022: [https://insurance.maryland.gov/Consumer/Appeals and Grievances Reports/2022-Report-on-the-Health-Care-Appeals-and-Grievance-Law.pdf](https://insurance.maryland.gov/Consumer/Appeals%20and%20Grievances%20Reports/2022-Report-on-the-Health-Care-Appeals-and-Grievance-Law.pdf)
2023: [https://insurance.maryland.gov/Consumer/Appeals and Grievances Reports/2023-Report-on-the-Health-Care-Appeals-and-Grievance-Law.pdf](https://insurance.maryland.gov/Consumer/Appeals%20and%20Grievances%20Reports/2023-Report-on-the-Health-Care-Appeals-and-Grievance-Law.pdf)
2024: [https://insurance.maryland.gov/Consumer/Appeals and Grievances Reports/Health-Care-Appeals-and-Grievance-Law-2024-Report-Insurance-Article-15-10A-06-b-3.pdf](https://insurance.maryland.gov/Consumer/Appeals%20and%20Grievances%20Reports/Health-Care-Appeals-and-Grievance-Law-2024-Report-Insurance-Article-15-10A-06-b-3.pdf)
2. I opened each PDF in Adobe Acrobat and deleted all pages that did not include the appendices.
3. I used Adobe Acrobat's built-in feature to convert each year's PDF into Excel format.
4. After conversion, the Excel files were not properly formatted. The columns were disorganized, and data spilled across multiple columns, which would make EDA messy and unstructured.

[illegible]

- ## Administration Complaints

6. I opened a second window of the same Excel workbook. One window displayed the original report, and the other displayed the appendix sheet I was working on.
7. I manually added column headers based on what was shown in the appendix.

8. I highlighted as many values as possible from the appendix (excluding column names) and pasted only the values into the corresponding sheet. I repeated this process until reaching the “Total” row, which indicates the end of the appendix.
9. I wrapped the company name’s column so they would remain within a single cell and clearly indicate where each column ends.

Worksheet looked like this

F6																											
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z		
company name	adverse decisions company total	adverse decisions by carrier percent	inpatient hospital services number	inpatient hospital services percent total	emergency room services number	emergency room services percent total	mental health services number	mental health services percent total	physician services number	physician services percent total	laboratory, radiology services number	laboratory, radiology services percent total	pharmacy services number	pharmacy services percent total	pt, et, st services (incl inpatient rehab) number	pt, et, st services (incl inpatient rehab) percent total	skilled nursing facility sub home number	skilled nursing facility sub acute nursing home percent total	skilled nursing facility sub acute nursing home percent total	skilled nursing facility sub acute nursing home percent total	skilled nursing facility sub acute nursing home percent total	skilled nursing facility sub acute nursing home percent total	skilled nursing facility sub acute nursing home percent total	skilled nursing facility sub acute nursing home percent total	skilled nursing facility sub acute nursing home percent total		
1	4 Ever Life Insurance Company																1										
2	Atma Dental, Inc.																693										
3	Atma Health Inc. (Pennsylvania corporation)																115										
4	Atma Life Insurance Company																188										
5	American Life Insurance Corp.																285										
6	CareFirst BlueChoice, Inc.																21128										
7	CareFirst of Maryland, Inc.																7057										
8	CGNA Dental Health of Maryland, Inc.																47										
9	CGNA Health and Life Insurance Company																17081										
10	Connecticut																7										

10. I selected all empty columns (from the first company name through the “Total” row), deleted them, and chose the “Shift cells left” option. I repeated this process until all columns were properly aligned.

One completed appendix looked like this:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA
	company name	adverse decisions by carrier total	adverse decisions by carrier percent	inpatient hospital services number	inpatient hospital services percent total	emergency room services number	emergency room services percent total	mental health services number	mental health services percent total	physician services number	physician services percent total	laboratory, radiology services number	laboratory, radiology services percent total	pharmacy services number	pharmacy services percent total	pt., et., st services (incl input rehab) number	pt., et., st services (incl input rehab) percent total	skilled nursing facility sub acute nursing home number	skilled nursing facility sub acute nursing home percent total	durable medical equipment services number	durable medical equipment services percent total	dental number	dental percent total	home health services number	home health services percent total	obesity, ivf, podiatry, hearing and vision number	obesity, ivf, podiatry, hearing and vision percent total
1	Blue Cross of Michigan	1	0	1	1.00	0	0.000	0	0.000	0	0.000	0	0.000	0	0	0	0.000	0	0.000	0	0.000	0	0.000	0	0.000	0	0.000
2	Aetna Dental, Inc.	993	0.009	0	0.00	0	0.000	0	0.000	0	0.000	0	0.000	0	0	0	0.000	0	0.000	0	0.000	693	1.000	0	0.000	0	0.000
3	Aetna Health Inc. (a Pennsylvania corporation)	115	0.001	39	0.34	0	0.000	4	0.035	42	0.365	2	0.017	15	0.13	4	0.035	2	0.017	2	0.017	0	0.000	0	0.000	5	0.043
4	Aetna Life Insurance Company	188	0.002	69	0.37	0	0.000	11	0.059	48	0.362	8	0.043	19	0.101	5	0.027	2	0.011	2	0.011	0	0.000	1	0.005	3	0.016
5	Anthem Life Insurance Company	285	0.004	0	0.00	0	0.000	0	0.000	0	0.000	0	0.000	0	0	0	0.000	0	0.000	0	0.000	285	1.000	0	0.000	0	0.000
6	Centene Corp.	21128	0.26	12	0.00	0	0.000	298	0.014	1182	0.056	3015	0.145	15727	0.744	85	0.004	6	0.000	786	0.037	4	0.000	9	0.000	4	0.000
7	Centene of Maryland, Inc.	7037	0.087	2	0.00	1	0.000	31	0.004	184	0.026	239	0.034	7810	0.025	14	0.002	2	0.000	111	0.016	659	0.093	4	0.001	0	0.000
8	CIGNA Dental Health of Maryland, Inc.	47	0.001	0	0.00	0	0.000	0	0.000	0	0.000	0	0.000	0	0	0	0.000	0	0.000	0	0.000	47	1.000	0	0.000	0	0.000
9	CIGNA Health and Life Insurance Company	17081	0.211	284	0.02	0	0.000	65	0.004	737	0.043	6699	0.392	5464	0.32	3499	0.205	3	0.000	48	0.003	162	0.009	85	0.005	35	0.002
10	Connecticut General Life Insurance Company	7	0	0	0.00	0	0.000	0	0.000	0	0.000	4	0.371	0	0	3	0.429	0	0.000	0	0.000	0	0.000	0	0.000	0	0.000
11	Delta Dental	8	0	0	0.00	0	0.000	0	0.000	0	0.000	0	0.000	0	0	0	0.000	0	0.000	0	0.000	8	1.000	0	0.000	0	0.000

11. I repeated this entire process for all years and appendices.

Merging All Datasets into One

To perform effective and efficient EDA, I consolidated all datasets into one complete, organized, and easy to understand dataset.

12. I created a new blank Excel workbook named “MIA Appeals and Grievances.”

13. I opened all yearly datasets.

14. I copied and pasted all values from the first appendix (2021) into the new workbook.

15. I repeated this process for all the years, pasting each year’s data below the previous one.

16. I added a Year column and filled in the appropriate year for each row.

17. I then copied the column headers from other appendices and pasted them into the main sheet. For service-related columns, I distinguished whether they represented adverse decisions or grievances.

18. I used Excel’s built-in XLOOKUP function to populate the dataset:

“=XLOOKUP(B2, '[MIA (YEAR).xlsx](Appendix Name)'!\$A:\$A'[MIA (YEAR).xlsx](Appendix Name)'!\$(Column):\$(Column), "N/A", 0)”

This formula searches the original appendix sheets by company name and fills in the corresponding data within the specified columns. If a company does not have data for a specific appendix, the value is represented with “N/A”.

19. I cross-checked the MIA Appeals and Grievances dataset with the original reports to ensure all values aligned correctly and that no companies were missing. This step was critical to ensure data accuracy.

20. I adjusted the formula for each year and appendix and repeated the process until all companies and years were fully populated

Completed MIA Appeals and Grievances:

Since one appendix breaks data down by procedure type, merging it into the main dataset was difficult due to the lack of a common key.

21. I created a separate Excel workbook called “Admin Complaints”

22. I copied and paste the appendix data by year into it

23. I added a Year column and filled in the appropriate year for each row.

Completed Admin Complaints:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
	year	Type of Procedure	complaint code	total	complaint upheld by MIA number	complaint upheld by mia percent	complaint reversed by mia number	complaint reversed by mia percent	complaint modified by mia number	complaint modified by mia percent	complaint reversed during investigation number	complaint reversed during investigation percent														
1																										
2	2021	Air Ambulance	B	2	2	1	0	0	0	0	0	0														
3	2021	Cosmetic	D	3	0	0	2	0.67	1	0.33	0	0														
4	2021	Dental Care	J	32	10	0.31	4	0.13	1	0.03	17	0.53														
		Durable Medical Equipment	I	7	2	0.29	3	0.43	0	0	2	0.29														
6	2021	Emergency	B	2	0	0	0	0	0	0	2	1														
6	2021	Room Denial																								
7	2021	Experimental	D	6	4	0.67	1	0.17	0	0	1	0.17														
8	2021	Eye Care	L	1	0	0	1	1	0	0	0	0														
		In-Patient Rehabilitation Services	G	2	2	1	0	0	0	0	0	0														
9	2021	Lab Imaging, Test Services	E	45	27	0.6	11	0.24	0	0	7	0.16														
10	2021	Laboratory Services	E	1	0	0	0	0	0	0	1	1														
11	2021	Mental Health/Substance (Inpatient) Services	C	2	0	0	0	0	0	0	2	1														
12	2021	Mental Health/Substance (Outpatient) Services	C	8	1	0.13	2	0.25	0	0	5	0.63														
13	2021	Opioid Use Disorders	F	1	1	1	0	0	0	0	0	0														
14	2021	PCP Referrals	D	1	0	0	0	0	0	0	1	1														
15	2021	Pharmacy Benefits	F	1	0	0	0	0	0	0	0	1														
16	2021	Pharmacy Services/Formulary Issues	F	157	31	0.2	37	0.24	1	0.01	88	0.56														
17	2021	Physician Services	D	35	9	0.26	6	0.17	2	0.06	17	0.49														
18	2021	Services																								
19	2021	PT, OT, ST	G	2	1	0.5	1	0.5	0	0	0	0														
20	2021	TOTAL		308	90	0.2925649	48	0.1558117	5	0.01623377	144	0.46753247														
21	2022	Cosmetic	D	3	2	0.67	1	0.33	0	0	0	0														

Final Data Preparation for R

24. I converted both the MIA Appeals and Grievances and Admin Complaints datasets into CSV format for use in RStudio.

25. I replaced all “N/A” values with blank cells so that RStudio would interpret them as null values rather than character strings.

26. After importing the data into RStudio, I reviewed the column names and noticed that they contained periods (adverse.decisions.company.total), which I did not prefer.

27. I used the following code to clean the column names:

```
“names(mia) <- tolower(names(mia))

names(mia) <- gsub(' ','_',names(mia))
```

```
names(mia) <- gsub("\\.", "_",names(mia))"
```

This code converts all column names to lowercase and replaces spaces and periods with underscores for consistency and readability.